Lesson notes · for you

Poisoned data

AI security · Upper secondary · Lesson 3

Time
20 min

Materials
Paper and a pen; Ten to twenty simple things to measure, such as leaves or beans, and a ruler; A phone or computer with Darsi, if you have one, for the lab and the code

Learning goals

- Explain data poisoning and backdoors.
- Find suspicious labels with a neighbour check in code.
- Decide what to remove, and record why.

Activity: Activity: Find the poison

1. Warm-up (3 min): Write down your guess: if two of twenty labels are wrong, will a model notice? Keep your guess to check later.
2. Measure (7 min): Measure ten things of two kinds (leaf or bean lengths) and label each one. Without looking, let a friend flip two of the labels.
3. Check (5 min): Sort the measurements from smallest to largest. Circle every label whose left and right neighbours both have the other label.
4. Share (5 min): Ask your friend which labels were flipped. Did you circle them? Did you circle an honest one at the edge of a group? Write down what you'd remove and why.

Measure harmless things, like leaves or beans. No device needed: the neighbour check works on paper.

Discussion

- Why can't a high test score prove the data is clean?
- Why should a person look before anything is removed?
- Where could someone plant a wrong label in your school's data?

Check yourself

- Explain the neighbour check in one sentence, and one thing it can get wrong.

Try it this week
With three friends, label the same ten pictures on your own, then compare. Where you disagree, talk it through and keep the majority, and notice which labels were careless.

Worksheet

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Name _____ Class _____ Date _____

1 **New data was added last night. This morning the score on your locked test set fell from 96% to 80%. What do you do first?**

◯ Train again with even more data ◯ Roll back, and check the new data

◯ Delete the test set: it must be wrong

2 **If a model scores well on all its normal tests, its training data can't have been poisoned.**

◯ True ◯ False

3 **Three friends label the same photo. What does this print?**

```
votes = ["sick", "sick", "ok"]
print(votes.count("sick") > len(votes) / 2)
```

◯ True ◯ False ◯ 2

4 **Finish suspects(xs, lab). The leaves are sorted by how yellow they are; lab holds their labels (H healthy, S sick). Return the yellow values whose label both neighbours disagree with. It should print [14, 74].**

```
def suspects(xs, lab):
    out = []
    for i in range(1, len(lab) - 1):
        left, me, right = lab[i-1:i+2]
        # flag xs[i] when both neighbours
        # disagree with me
        pass
    return out

yellow = [10, 12, 14, 16, 18,
          70, 72, 74, 76, 78]
lab = "HHSHHSSHSS"
print(suspects(yellow, lab))
```

Write your program:

5 **This version of suspects flags far too many leaves, including honest ones at the edge of each group. Which line has the mistake?**

Circle the line with the mistake. Write it correctly:

```
def suspects(xs, lab):
    out = []
    for i in range(1, len(lab) - 1):
        left, me, right = lab[i-1:i+2]
        if left != me or right != me:
            out.append(xs[i])
    return out
```

✎ **Think of data you or your class have collected: a survey, measurements, photos. How could a wrong or planted entry get in, and how would you spot it?**
